

Ph.D. Thesis Defense

Generalized t-SNE and Beyond: Probabilistic Methods for Dimensionality Reduction, Combinatorial Optimization, and Machine Learning

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This thesis defense is about probabilistic algorithms along with the problems they aim to tackle.

- Two basic problems: Dimensionality and Dependence
- Dimensionality Reduction and t-SNE
- Dependence, Probabilistic Graphic Models and Lovász Local Lemma
- Autoencoder and Variational Factor Analysis

Dimensionality has been with us since the dawn of time. It refers to different aspects of the information. The scale of dimensionality drastically increases during the modern era because of advancements in mathematics and computer science.

- Increasing dimensionality means we can store more data, even the raw information of the process we aim to analyze.
- However, it does not mean we are automatically able to extract insights from the data.

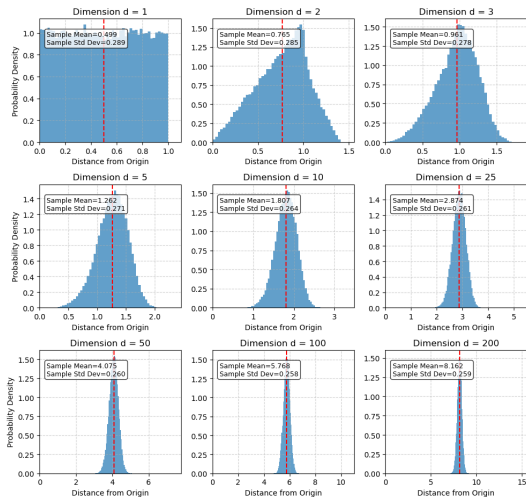
The curse of dimensionality: a series of phenomena that arises in high-dimensional spaces that are usually not present or prominent in low-dimensional settings.

Example: The volume of a hypercube gets concentrated in its corners. Let C_d , S_d be the volume of the hypercube and hypersphere respectively.

$$\lim_{d \rightarrow \infty} \frac{S_d}{C_d} = \lim_{d \rightarrow \infty} \frac{\pi^{d/2}}{d 2^{d-1} \Gamma(d/2)} = 0. \quad (1)$$

Curse of dimensionality (cont.)

Example: the distribution of distance between uniformly distributed points is concentrated in high-dimensional spaces.



t-distributed Stochastic Neighbor Embedding (t-SNE) is an algorithm that find a low-dimensional representation of high-dimensional data.

$d, s \geq 2$ with $d > s$, and $n \geq 1$.

- **Input:** $X_1, \dots, X_n \in \mathbb{R}^d$ the original data points.
- **Output:** $Y_1, \dots, Y_n \in \mathbb{R}^s$ the low-dimensional representation.

Overall idea: KL-divergence (relative entropy) can be used to measure the difference between two distributions. t-SNE construct two distributions from both input and initial output data points, then minimize the KL-divergence by adjusting the output.

The concept of kernels

Definition (Generalized Input Kernel)

The input kernel $\mathbf{Ker_I}(x, y, \sigma) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^+$ is a function of the input data points and parameter σ such that for any $x, y \in \mathbb{R}^d$ and $\sigma > 0$,

$$\mathbf{Ker_I}(x, y, \sigma) = \mathbf{Ker_I}(y, x, \sigma). \quad (2)$$

Definition (Generalized Output Kernel)

The output kernel $\mathbf{Ker_O}(z, w) : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}^+$ is a function of the output data points such that for any $z, w \in \mathbb{R}^s$,

$$\mathbf{Ker_O}(z, w) = \mathbf{Ker_O}(w, z). \quad (3)$$

The concept of kernels(cont.)

Construct probability distributions from the input kernels:

$$p_{j|i} = \frac{\mathbf{Ker_I}(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker_I}(X_i, X_k, \sigma_i)}, \quad p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}. \quad (4)$$

For output kernels:

$$q_{ij} = \frac{\mathbf{Ker_O}(Y_i, Y_j)}{\sum_{k \neq l} \mathbf{Ker_O}(Y_k, Y_l)}. \quad (5)$$

Original t-SNE can be recovered by letting

$$\mathbf{Ker_I}(x, y, \sigma) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right), \quad \mathbf{Ker_O}(z, w) = \frac{1}{1 + \|z - w\|^2}. \quad (6)$$

Recall

$$p_{j|i} = \frac{\mathbf{Ker_I}(X_i, X_j, \sigma_i)}{\sum_{k \neq i} \mathbf{Ker_I}(X_i, X_k, \sigma_i)}. \quad (7)$$

σ_i satisfies that for each $i = 1, \dots, n$,

$$- \sum_{j=1, j \neq i}^n p_{j|i} \log p_{j|i} = \log \mathbf{Perp} =: \log(n\rho), \rho \in (0, 1). \quad (8)$$

Intuition: σ_i is the zoom slide bar for the input data points.

- **Input:** $X_1, \dots, X_n \in \mathbb{R}^d \implies p_{ij}$.
- **Output:** $Y_1, \dots, Y_n \in \mathbb{R}^s \implies q_{ij}$.
- **Variables Involved:** p_{ij} are fixed. q_{ij} can be changed and is a function of Y_i and Y_j .
- **Loss Function (Goal):** minimize the KL-divergence:

$$L_{n,\rho}(X, Y) = \sum_{1 \leq i \neq j \leq n} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right). \quad (9)$$

- **End Result:**

$$Y^* = \arg \min_{Y \in (\mathbb{R}^s)^n} L_n(X, Y). \quad (10)$$

Continuous version of the entropy:

$$F_{\rho,\mu}(x, \sigma) = \int \frac{\mathbf{Ker_I}(x, x', \sigma)}{\int \mathbf{Ker_I}(x, x', \sigma) d\mu(x')} \log \left(\frac{\mathbf{Ker_I}(x, x', \sigma)}{\int \mathbf{Ker_I}(x, x', \sigma) d\mu(x')} \right) d\mu(x') + \log \rho. \quad (11)$$

$\sigma_{\rho,\mu}^* : \mathbb{R}^d \rightarrow \mathbb{R}$ the unique solution of the equation

$$F_{\rho,\mu}(x, \sigma) = 0. \quad (12)$$

σ_i serves the anchor point of distributions induced by the input data. $\sigma_{\rho,\mu}^*$ is its asymptotic limit.

Continuous version of induced probability distributions:

$\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ be a given function and $p_\psi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be defined by

$$p_\psi(x, x') = \frac{1}{2} \left(\frac{\mathbf{Ker_I}(x, x', \psi(x))}{\int \mathbf{Ker_I}(x, x', \psi(x)) d\mu(x')} + \frac{\mathbf{Ker_I}(x, x', \psi(x'))}{\int \mathbf{Ker_I}(x, x', \psi(x')) d\mu(x)} \right). \quad (13)$$

$q : \mathbb{R}^s \times \mathbb{R}^s \rightarrow \mathbb{R}$ be given by

$$q(y, y') = \frac{\mathbf{Ker_O}(y, y')}{\iint \mathbf{Ker_O}(y, y') d\mu(y) d\mu(y')}. \quad (14)$$

Continuous version of the loss function:

$$I_\rho(\mu) = \iint p_{\sigma_{\rho,\mu}^*}(x, x') \log \left(\frac{p_{\sigma_{\rho,\mu}^*}(x, x')}{q(y, y')} \right) d\mu(x, y) d\mu(x', y'). \quad (15)$$

Recall $\mu_X \in \mathcal{M}(\mathbb{R}^d)$ is an arbitrary given probability measure. We set $\mathcal{P}_X$ to be the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^s$ whose marginal coincides with μ_X . Formally speaking,

$$\mathcal{P}_X := \{\mu \in \mathcal{M}(\mathbb{R}^d \times \mathbb{R}^s) : \text{the } \mathbb{R}^d \text{ marginal of } \mu \text{ is } \mu_X\}. \quad (16)$$

Lastly, define

$$\tilde{\mathcal{P}}_X := \left\{ \mu \in \mathcal{P}_X : \int (p_{\sigma_{\rho,\mu}^*}(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} \mathbf{Ker}_O(y, y')}{\mathbf{Ker}_O(y, y')} d\mu(x', y') = 0, \mu(x, y)\text{-a.e.} \right\}. \quad (17)$$

Condition (Input Kernel Conditions)

Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be a fixed function. The input kernel has the form of

$$\mathbf{Ker_I}(x, x', \sigma) = \exp(-w(\sigma \|x - x'\|^\theta)) \quad (18)$$

Here, $\theta > 0$ is a fixed constant. w satisfies:

1. $\forall t \geq 0, w'(t) > 0. \lim_{t \rightarrow \infty} w(t) = \infty.$
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0.$
3. The following integral converges:

$$\int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (19)$$

Since the all probability masses are normalized, we can assume without loss of generality that $w(0) = 0$ from now on.

Condition (Output Kernel Conditions)

The output kernel depends only on the distance between two output data points y and y' . In particular, $\mathbf{Ker}_O(y, y') = k(\|y - y'\|)$ and:

- 1. $k : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^+$ is decreasing and bounded.*
- 2. $\mathbf{Ker}_O$ satisfies*

$$\iint_{(0, \infty)^2} \mathbf{Ker}_O(y, y') dy dy' < \infty. \quad (20)$$

- 3. k has bounded derivative.*
- 4. $k'(0) = 0$.*

Theorem

Assume μ_X has a continuous density $f(x)$ and compact support. Let $X_i \in \mathbb{R}^d$ be i.i.d random variables with probability measure μ_X . If both input and output condition hold, then for any $\rho \in (0, 1)$,

$$\lim_{n \rightarrow \infty} \inf_Y L_{n,\rho}(X, Y) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu). \quad (21)$$

Moreover, there exist a measure μ^ and a sub-sequence $\{n_k\}$ such that $\mu_{n_k} \rightarrow \mu^*$ weakly and $I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu)$.*

Theorem

Under the same assumptions of theorem 3, μ^ has compact support.*

Are the conditions we impose on input and output kernels general enough?

- t-SNE is about the relationship between the data points. No generality loss by assuming the kernels are functions of the distance.
- Kernels are always non-negative. No generality loss by inserting exponential functions in input kernels for it being a smooth bijection.
- σ is only used to control entropy (and in our case it does).
- Integrability is the bare minimum.
- The rest is only about decaying speed.

The problems of dependence

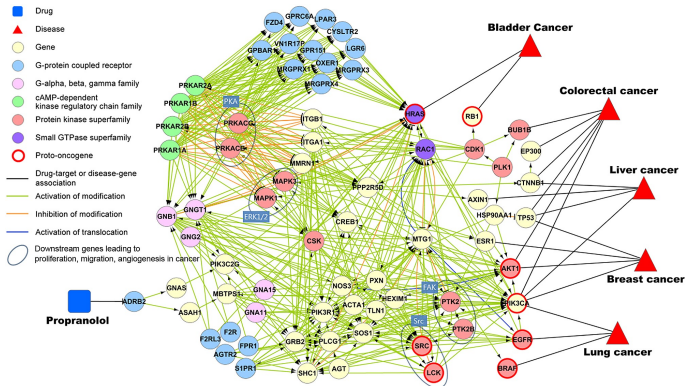
Textbooks start with independence, real problems strand because of dependence.

- Weather, genetic expression, finance, etc.
- For math itself, too many theorems cannot be extended beyond Gaussian or i.i.d. settings.

Probabilistic Graphical Models

Probabilistic graphical models (PGMs) are a class of statistical models that use graphs to represent the dependencies among random variables.

Example: genetic expression:



Let $[n] = \{1, 2, \dots, n\}$ and $\mathcal{X} = (X_t)_{t \in [n]}$ be a finite dimension random vector with each X_t taking values in a finite set $\mathcal{S}_t = \{1, 2, \dots, m_t\}$. Suppose $[n]$ is the set of vertices of a graph. For each t , denote $\mathcal{N}_t$ the set of vertices s such that s is adjacent to t in the graph.

Definition

The process $(X_t)_{t \in [n]}$ is said to be a Markov random field if

(i) for each tuple $x = (x_1, x_2, \dots, x_n) \in \prod_{i=1}^n \mathcal{S}_i$, the joint probability satisfies

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) > 0; \quad (22)$$

(ii) for each t and x ,

$$\mathbb{P}\{X_t = x_t | X_s = x_s, s \neq t\} = \mathbb{P}\{X_t = x_t | X_s = x_s, s \in \mathcal{N}_t\}. \quad (23)$$

A subset of vertices $A \subset [n]$ is said to be complete (or a clique) if each pair of vertices is connected with an edge in the graph.

Definition

Let $x = (x_1, x_2, \dots, x_n) \in \prod_{i=1}^n \mathcal{S}_i$ be an arbitrary tuple from the joint distribution, and $\mathcal{C}$ the collection of all complete vertex subsets. A probability distribution Q is called a Gibbs distribution (or Gibbs random field) if it can be written in the form

$$Q(x) = \prod_{A \in \mathcal{C}} V_A(x), \quad (24)$$

where $V_A(x)$ is a positive function that depends only on x_i 's with $i \in A$.

Hammersley-Clifford theorem

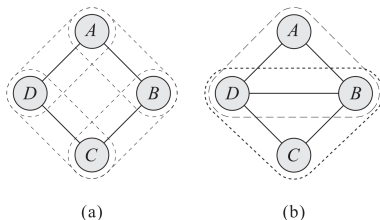
It is clear that each process $(X_t)_{t \in [n]}$ has a corresponding Gibbs distribution based on the dependencies. Now we can state the powerful Hammersley-Clifford theorem:

Theorem (Hammersley-Clifford theorem)

Let Q be a probability distribution on $\prod_{i=1}^n \mathcal{S}_i$. Then Q is a Gibbs distribution if and only if it is a Markov random field with respect to the same graph structure.

Alternative formulation of MRFs

Example:



In the two graphs above, the factor of the value assignments are

$$\begin{aligned}\tilde{p}(a, b, c, d) &= \phi(a, b)\phi(b, c)\phi(c, d)\phi(a, d), \\ \tilde{p}(a, b, c, d) &= \phi(a, b, d)\phi(b, c, d).\end{aligned}\tag{25}$$

where ϕ can be any positive functions defined by the user.

$$p(a, b, c, d) = \frac{1}{Z} \tilde{p}(a, b, c, d), \quad Z = \sum_{(a,b,c,d)} \tilde{p}(a, b, c, d).\tag{26}$$

Definition (Alternative Definition of MRF)

Let $X_1, \dots, X_n$ be random variables. A Markov Random Field (MRF) is the joint probability distribution of $(X_i)_{i \in [n]}$ over an undirected graph $G = (V := [n], E)$ with each vertex i representing the random variable X_i . In addition, the joint probability p has the form

$$p(x_1, \dots, x_n) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \phi_C(x_C), \quad (27)$$

where $\mathcal{C}$ is the collection of all cliques (complete subgraphs) in G , each factor ϕ_C is a non-negative function, and

$$Z = \sum_{x_1, \dots, x_n} \prod_{C \in \mathcal{C}} \phi_C(x_C) \quad (28)$$

is the normalizing partition function.

Exponential family and constrained MRFs

We can define MRF as a Boltzmann distribution of the discrete variables $X = \{X_i\}_{i=1}^n$ over a Boolean hypercube $\mathcal{X} = \{0, 1\}^n$. For $x \in \mathcal{X}$, we have:

$$P_{\theta}(X = x) = \frac{\exp(\phi_{\theta}(x))}{Z(\theta)}. \quad (29)$$

Here, $Z(\theta) = \sum_{x' \in \mathcal{X}} \exp(\phi_{\theta}(x'))$ is the partition function.

$\phi_{\theta}(x) = \sum_{j=1}^m \phi_{\theta,j}(x_j)$ is the potential function. Each $\phi_{\theta,j}$ is a *factor potential*, which maps a value assignment over a subset of variables $X_j \subseteq X$ to a real number. For a set of hard constraints $\mathcal{C} = \{c_j\}_{j=1}^L$. Define the constrained MRF as

$$P_{\theta}(X = x | \mathcal{C}) = \frac{\exp(\phi_{\theta}(x)) C(x)}{Z_{\mathcal{C}}(\theta)}, \quad (30)$$

where $C(x) = \prod_{j=1}^L c_j(x)$ is the indicator function.

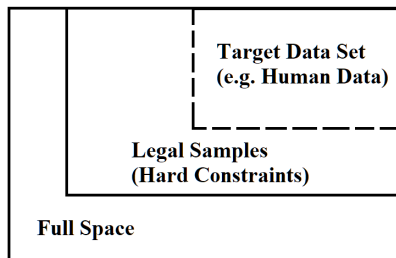
MRF can be used to represent constraint-satisfaction problems where each vertex represents a constraint and an edge is drawn if two constraints share at least one variable.

Definition (Extremal Condition)

A set of constraints $\mathcal{C}$ is called extremal if and only if for each pair of constraints $c_i, c_j \in \mathcal{C}$, (i) either their domain variables do not intersect, i.e., $\text{var}(c_i) \cap \text{var}(c_j) = \emptyset$. (ii) or for all $x \in \mathcal{X}$, $c_i(x) = 1$ or $c_j(x) = 1$.

The extremal condition represents a wide class of constraints. For constraints that can fail simultaneously, we can choose a smaller data space to make the setup more concise and achieve extremal conditions.

The context of machine learning problems (cont.)



The computation of negative log-likelihood's gradient is computationally intractable because of the partition function $Z(\theta)$ and corresponding conditional distribution.

$$\begin{aligned}\nabla \ell_{\mathcal{C}}(\theta) &= -\frac{1}{N} \sum_{k=1}^N \nabla \phi_{\theta}(x^k) + \nabla \log Z_{\mathcal{C}}(\theta) \\ &= -\mathbb{E}_{x \sim \mathcal{D}} (\nabla \phi_{\theta}(x)) + \mathbb{E}_{\tilde{x} \sim P_{\theta}(x|\mathcal{C})} (\nabla \phi_{\theta}(\tilde{x})) .\end{aligned}\tag{31}$$

In order to compute the term $\mathbb{E}_{\tilde{x} \sim P_\theta(x|\mathcal{C})} (\nabla \phi_\theta(\tilde{x}))$ using Monte Carlo sampling, we need to sample from the distribution $P_\theta(x|\mathcal{C})$.

Lemma (Original Lovász Local Lemma (1975))

Let $A_1, \dots, A_k$ be a sequence of events such that each event occurs with probability at most p and such that each event is independent of all the other events except for at most d of them. Assume

$$4pd \leq 1. \tag{32}$$

Then there is a nonzero probability that none of the events occurs.

Sampling algorithm using Lovász Local Lemma

Input: Random variables $X = \{X_i\}_{i=1}^n$; Constraints $\mathcal{C} = \{c_j\}_{j=1}^L$; Initial parameters of the constrained MRF θ .

$x_i \sim \frac{\exp(\theta_i x_i)}{\sum_{x_i \in \{0,1\}} \exp(\theta_i x_i)}$, for $1 \leq i \leq n$. ▷ initialize

while $C(x) = 0$ **do**

Find all violated constraints $S \subseteq \mathcal{C}$ in x .

$x_k \sim \frac{\exp(\theta_k x_k)}{\sum_{x_k \in \{0,1\}} \exp(\theta_k x_k)}$, for $x_k \in \text{var}(S)$. ▷ resample

end while

return A valid sample x drawn from $P_\theta(X = x|\mathcal{C})$.

Theoretical guarantees of the sampling algorithm

The sampling output follows the exact conditional distribution.

Theorem (Probability Distribution)

Given random variables $X = \{X_i\}_{i=1}^n$, constraints $\mathcal{C} = \{c_j\}_{j=1}^L$ that satisfy the extremal condition, and the parameters of the constrained MRF in the single variable form θ . Upon termination, the sampling algorithm outputs an assignment x that is randomly drawn from the constrained MRF distribution: $x \sim P_\theta(X = x|\mathcal{C})$.

Running time estimation

Let T_i be the total number of samplings that are related to the i -th constraint c_i (meaning the resampling changes variables related to c_i). Our work gave a new, correct proof of the following lemma and theorem.

Lemma (Guo, 2017 and Gu et al., 2023)

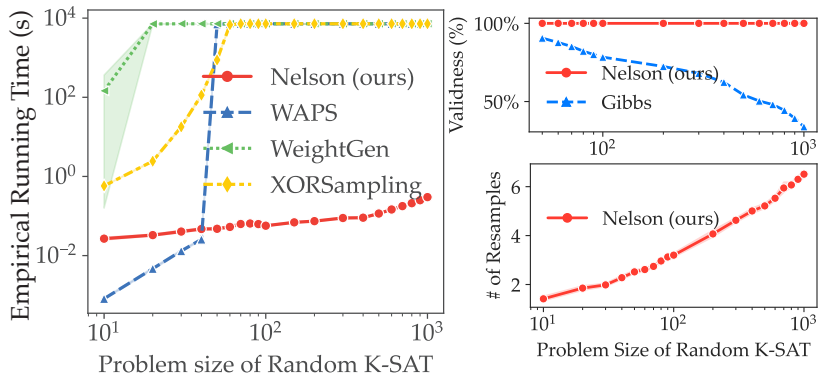
Let $q_\emptyset$ be a non-zero probability of all the constraints are satisfied. Let $q_{\{c_j\}}$ denote the probability that only constraint c_j is broken and the rest all hold. If $q_\emptyset > 0$, then $\mathbb{E}(T_i) = q_{\{c_j\}}/q_\emptyset$.

Theorem (Guo, 2017 and Gu et al., 2023)

Under extremal condition, the total number of re-samplings throughout the algorithm is then $\frac{1}{q_\emptyset} \sum_{j=1}^L q_{\{c_j\}}$.

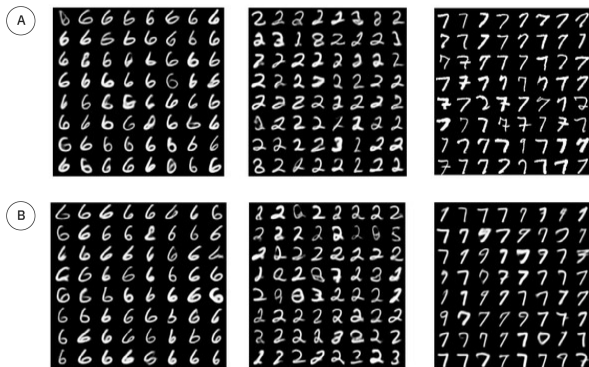
Selected experimental results: random K-SAT

Our ML framework based on LLL and MRF outputs 100% valid samples with unprecedented efficiency. The strength resides in the underlying probabilistic graphical model and the sampling algorithm.

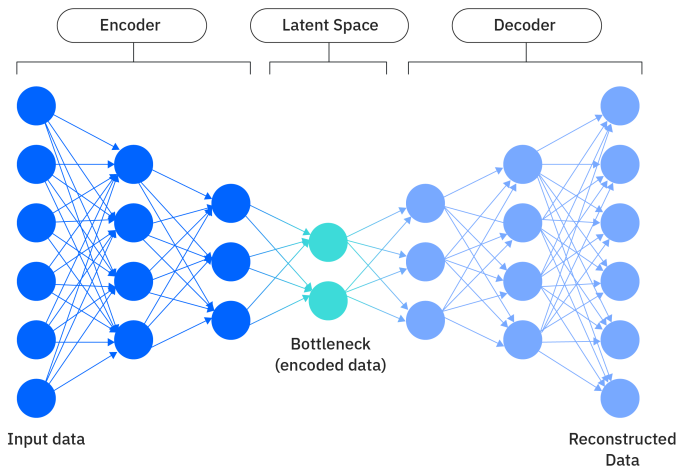


The concept of latent dimensions

Actual dimensions (all pixel data) vs latent dimension (which digit to draw):



Autoencoder: combining PGMs and dimensionality reduction



Assume there are p rounds of observations for n individuals. FA assumes that the collective behavior of the individuals can be explained by k common factors, and $k < p$. Mathematically speaking,

$$x_{i,m} - \mu_i = \sum_{j=1}^k l_{i,j} f_{j,m} + \epsilon_{i,m}. \quad (33)$$

Here, $x_{i,m}$ is the i -th observation of the m -th individual, μ_i is the mean of the i -th observation, $l_{i,j}$ is the loading of the j -th factor on the i -th observation, $f_{j,m}$ is the j -th factor score for the m -th individual and $\epsilon_{i,m}$ is the error term, usually with zero mean and finite variance.

Combining factor analysis with deep neural network gives us interpretable generative models.